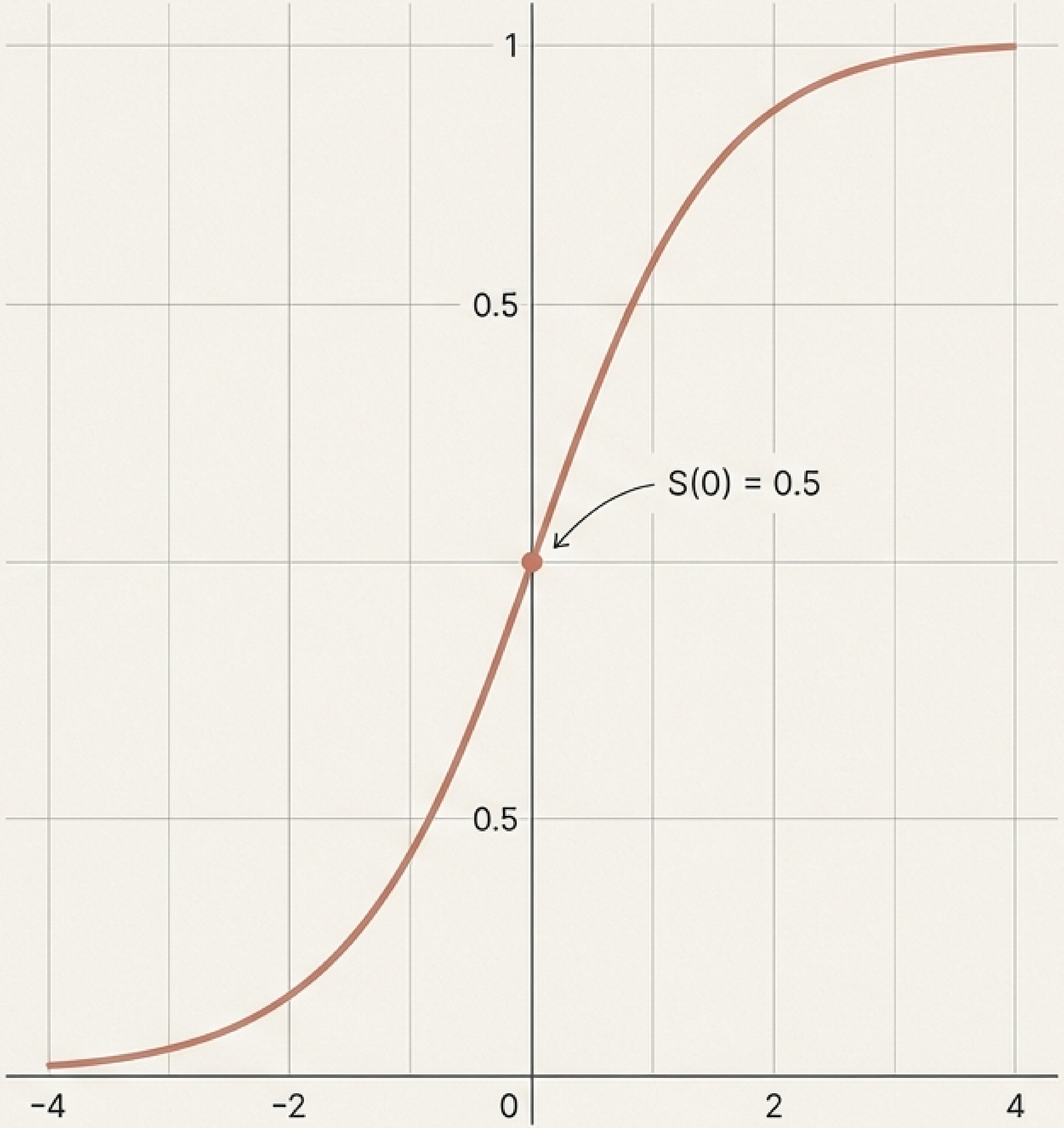
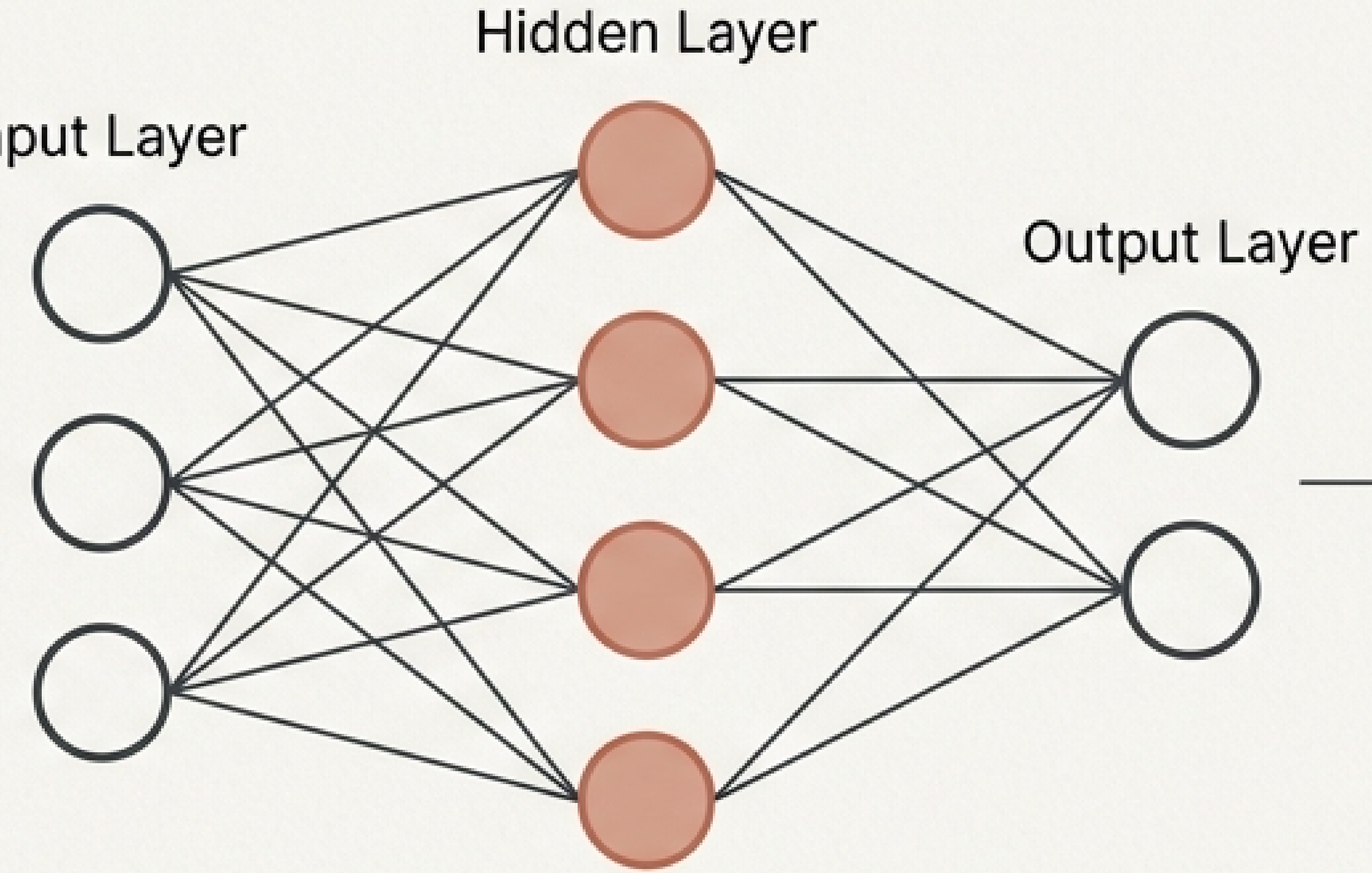
Deriving the Sigmoid Function

A step-by-step mathematical proof for Machine Learning applications.



Why the Sigmoid derivative matters in Machine Learning

Essential for Logistic Regression, Perceptrons, and Multi-Layer Perceptrons.



Used heavily in loss functions.

To optimise these networks, we must calculate gradients (slopes).
This requires knowing the exact derivative of the Sigmoid function.

Meet the Sigmoid function

$$\sigma(x) = \frac{1}{\underbrace{1 + e^{-x}}_{\text{Squashes any number—whether a massive positive or deep negative—into a strict range between 0 and 1.}}}$$

NotebookLM

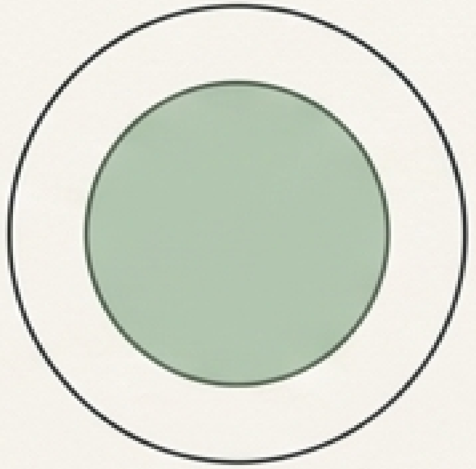
The mathematical toolkit required for the derivation

The Reciprocal Rule

$$x^{-1} \to -x^{-2}$$

Moving a denominator to the numerator creates a negative exponent. Annotatar in CMU Serif.

The Chain Rule



Required for differentiating nested functions. Differentiate the outside, then multiply by the derivative of the inside. Annotation in Inter.

Step 1: Restructuring for easier differentiation

$$\frac{d}{dx}\left[\frac{1}{1+e^{-x}}\right] \longrightarrow \frac{d}{dx}\left[(1+e^{-x})^{-1}\right]$$

Instead of using the complex quotient rule, we treat the entire denominator as a single term and raise it to the power of -1.

Step 2: Applying the reciprocal rule

The power of -1 drops to the front.

We subtract 1 from the exponent, resulting in -2.

$$-1 \cdot (1 + e^{-x})^{-2}$$

Step 3: Applying the chain rule to the inner function

$$-1 \cdot (1 + e^{-x})^{-2} \cdot \underbrace{\frac{d}{dx}(1 + e^{-x})}$$

- The derivative of the constant 1 is 0.
- The derivative of e^{-x} is e^{-x} multiplied by the derivative of its own exponent (-x).

Step 4: Resolving the chain rule and cancelling negatives

$$\cancel{-1} \cdot (1 + e^{-x})^{-2} \cdot e^{-x} \cdot (\cancel{-1})$$

The negative from the reciprocal rule and the negative from differentiating -x cancel each other out perfectly.

$$\frac{e^{-x}}{(1 + e^{-x})^2}$$

Step 5: Splitting the fraction to reveal a pattern

$$\frac{e^{-x}}{(1+e^{-x})^2} \longrightarrow \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}}$$

By pulling apart the squared denominator, we isolate two separate terms. This reveals a highly familiar component.

Step 6: Spotting the original Sigmoid function

$$\boxed{\sigma(x)} \cdot \frac{e^{-x}}{1 + e^{-x}}$$

The first term is exactly our original Sigmoid function. We can substitute σ(x) straight back into the equation.

Step 7: The '+1 -1' mathematical manipulation

$$\sigma(x) \cdot \frac{\underline{1 +} \, e^{-x} \, \underline{- 1}}{1 + e^{-x}}$$

We need a slight manipulation for the second term. Adding and subtracting 1 in the numerator changes nothing computationally, but it unlocks the final step.

Step 8: Finalising the algebraic substitution

$$\sigma(x) \cdot \left(\frac{1 + e^{-x}}{1 + e^{-x}} - \frac{1}{1 + e^{-x}} \right)$$

Any term divided by itself is 1.

This is the Sigmoid function again: $\sigma(x)$.

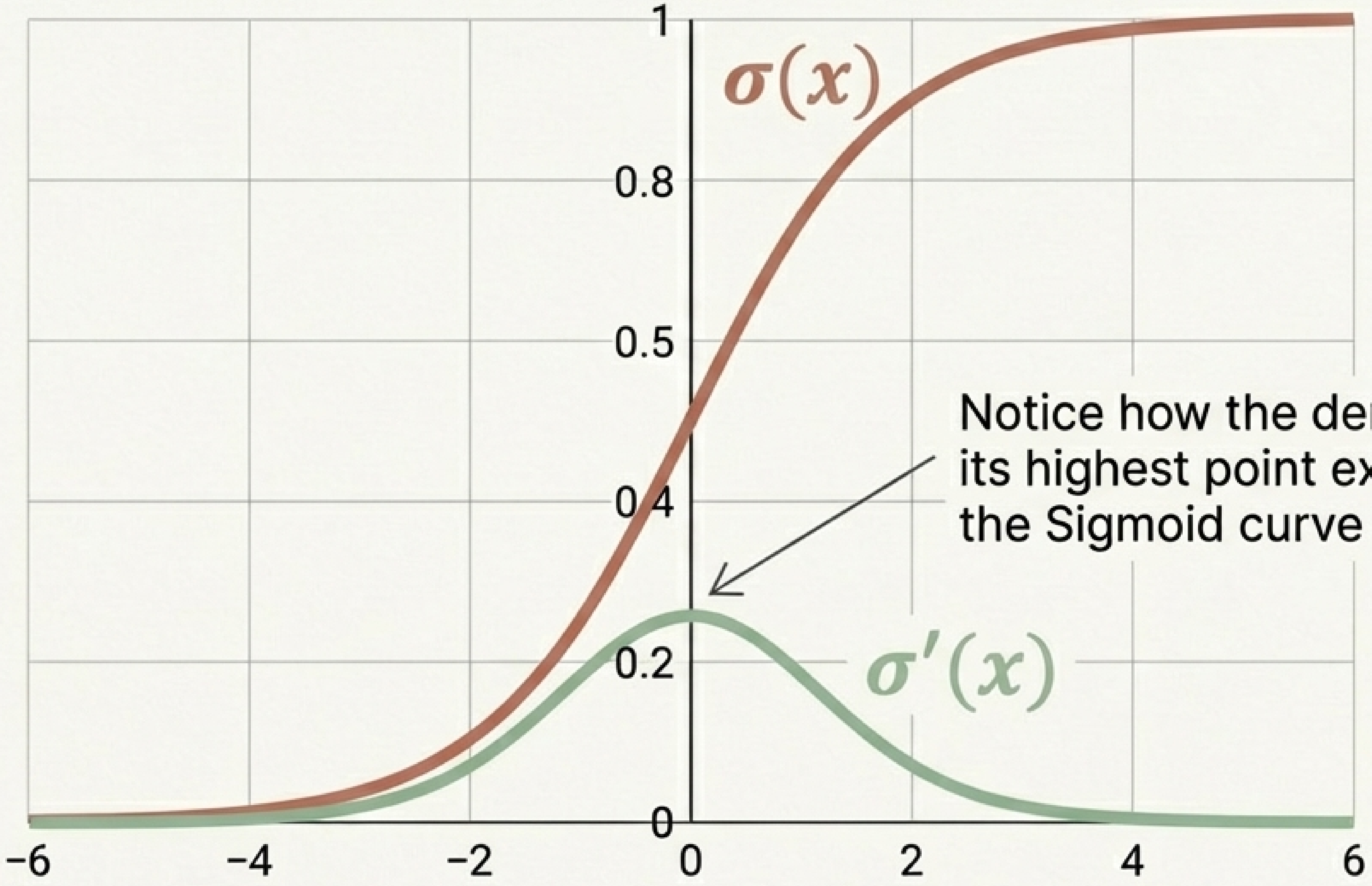
$$\sigma(x)(1 - \sigma(x))$$

The elegant final derivative

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

- Functions whose derivatives contain themselves are rare.
- Because the derivative is expressed entirely using the original function, calculating gradients in machine learning algorithms becomes highly computationally efficient.

Visualising the Sigmoid and its derivative



Notice how the derivative is at its highest point exactly where the Sigmoid curve is steepest.

Key takeaways for Machine Learning practitioners

- Essential for ML models—particularly perceptrons and logistic regression—to squash outputs safely between 0 and 1.
- The derivation relies on combining the reciprocal rule, the chain rule, and a clever '+1 −1' algebraic trick.
- The uniquely self-referential final formula, $\sigma(x)(1 - \sigma(x))$, ensures rapid, efficient gradient calculations during backpropagation.